Average Rate of Change

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing average rate of change**

Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2]$ $= 2$ **b** $[1, 3]$ $= 4$ **c** $[2, 5]$ $= 7$ **d** $[-1, 1]$ $= 0$

(a) $= 2$; (b) $= 4$; (c) $= 7$; (d) $= 0$

2 **[Foundational]** • **Target: Computing average rate of change**

Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.

$\frac{f(3) - f(0)}{3 - 0} = \frac{8 - 1}{3} = \frac{7}{3}$.

3 **[Foundational]** • **Target: Computing average rate of change**

Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

$\frac{g(b) - g(a)}{b - a} = \frac{(3b - 5) - (3a - 5)}{b - a} = \frac{3(b - a)}{b - a} = 3$. A linear function changes at a constant rate equal to its slope.

4 **[Foundational]** • **Target: Applications**

A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

a Find the average rate of change of the height over $[0, 2]$. $\frac{h(2) - h(0)}{2 - 0} = \frac{20 - 0}{2} = 10$ m/s.

b Find the average rate of change of the height over $[1, 3]$. $h(1) = 15$ and $h(3) = 15$, so the average rate of change is $\frac{15 - 15}{2} = 0$ m/s.

c Interpret your answer to part b in the context of the ball's motion. Over $[1, 3]$ the ball rises and then falls back to the same height, so on average its height does not change.

(a) $\frac{h(2) - h(0)}{2 - 0} = \frac{20 - 0}{2} = 10$ m/s.; (b) $h(1) = 15$ and $h(3) = 15$, so the average rate of change is $\frac{15 - 15}{2} = 0$ m/s.; (c) Over $[1, 3]$ the ball rises and then falls back to the same height, so on average its height does not change.

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5 [Foundational] • Target: Applications

The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.

$f(0) = 1$ and $f(4) = -8 + 8 + 1 = 1$, so the average rate of change over $[0, 4]$ is $\frac{1-1}{4} = 0$. The secant line through the endpoints is horizontal.

6 [Foundational] • Target: Applications

A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.

a Find the average rate of change of revenue over $[0, 4]$. $R(0) = 40$, $R(4) = 40 + 24 - 8 = 56$: rate $= \frac{56-40}{4} = 4$ thousand riyals/year.

b Find the average rate of change of revenue over $[4, 8]$. $R(8) = 40 + 48 - 32 = 56$: rate $= \frac{56-56}{4} = 0$ thousand riyals/year.

c Compare your two answers and interpret what they say about the company's growth. Revenue grew steadily during the first period but plateaued during the second — the average growth rate dropped to zero.

(a) $R(0) = 40$, $R(4) = 40 + 24 - 8 = 56$: rate $= \frac{56-40}{4} = 4$ thousand riyals/year.; (b) $R(8) = 40 + 48 - 32 = 56$: rate $= \frac{56-56}{4} = 0$ thousand riyals/year.; (c) Revenue grew steadily during the first period but plateaued during the second — the average growth rate dropped to zero.

7 [Foundational] • Target: Comparing rates and estimating from a table

A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3]$ $= \frac{17-5}{2} = 6$ **b** $[3, 6]$ $= \frac{44-17}{3} = 9$ **c** $[6, 10]$ $= \frac{92-44}{4} = 12$

(a) $= \frac{17-5}{2} = 6$; (b) $= \frac{44-17}{3} = 9$; (c) $= \frac{92-44}{4} = 12$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Comparing rates and estimating from a table

Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g's graph?

The average rate of change is increasing $(6, 9, 12)$, suggesting g is concave up — consistent with g being a quadratic or otherwise increasingly steep function.

9 [Intermediate] • Target: Comparing rates and estimating from a table

Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

d_1: $\frac{30-10}{4} = 5$ km/h. d_2: $\frac{36-4}{4} = 8$ km/h. The second traveler had the greater average speed over this interval.

10 [Intermediate] • Target: Average rate of change and secant lines

Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

The average rate of change equals the slope of the secant line connecting the two points $(a, f(a))$ and $(b, f(b))$ on the graph of f.

11 [Intermediate] • Target: Average rate of change and secant lines

For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.

$f(-1) = -1$, $f(2) = 8$. Rate $= \frac{8-(-1)}{2-(-1)} = \frac{9}{3} = 3$. Secant line: $y - 8 = 3(x - 2)$, i.e. $y = 3x + 2$.

12 [Intermediate] • Target: Comparing average rate of change across different functions

Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

f: $\frac{16-1}{4} = 3.75$. g: $\frac{16-0}{4} = 4$. Over this interval, $g(x) = x^2$ had the slightly larger average rate of change.

13 [Intermediate] • Target: Comparing average rate of change across different functions

Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

$f(4) = 16$, $f(8) = 256$. Rate $= \frac{256-16}{4} = 60$. This is much larger than the rate over $[0, 4]$ (3.75), illustrating that exponential functions grow at an accelerating average rate over successive equal-width intervals, unlike linear functions.

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14 [Intermediate] • Target: Average rate of change from a graph

Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.

From the graph: $f(0) = 3$, $f(4) = 3$. Average rate of change $= \frac{3-3}{4-0} = 0$, matching the fact that the parabola's vertex lies at the midpoint $x = 2$ and the two endpoints have equal height.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

$\frac{V(5) - V(0)}{5 - 0} = \frac{125 - 200}{5} = -15$ liters/minute. The negative sign means the tank is draining on average at 15 liters per minute over this interval.

16 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 6.02 Polar Equations and Graphs

A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.

$P(0) = 500$, $P(10) = 500e^1 \approx 1359$. Average rate of change $\approx \frac{1359 - 500}{10} \approx 86$ people/year.

17 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

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18 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 7.02 Logarithms and Their Properties

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?

We can use the Law of Cosines to solve for the unknown side BC. The Law of Cosines states that for any triangle with sides a, b, and c, and opposite angles A, B, and C respectively: $a^2 = b^2 + c^2 - 2bc\cos(A)$. In this problem, we have: side b = AC = 10, side c = AB = 14, and angle A = 52 degrees. We want to find side a = BC. Substituting the values into the formula: $a^2 = 10^2 + 14^2 - 2(10)(14)\cos(52^\circ)$ $a^2 = 100 + 196 - 280\cos(52^\circ)$ $a^2 = 296 - 280(0.61566)$ $a^2 = 296 - 172.3848$.

19 [Challenging] • Target: Formula and Secant Line Concept • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?

We have a circle with radius $r = 6$ and a sector with a central angle of 60 degrees. The chord connecting the endpoints of the arc of this sector forms an isosceles triangle with the two radii. Since the central angle is 60 degrees and the other two angles in the triangle must be equal (because the sides opposite them are radii), each of the other angles is $(180^\circ - 60^\circ)/2 = 60^\circ$. Therefore, the triangle is equilateral. In an equilateral triangle, all sides are equal. Since two sides are radii of.

20 [Challenging] • Target: Tabular and Data-Based Rates of Change • Cross-Lesson: 7.04 Applications: Growth and Decay

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?

This is a right-angled triangle, so we can use the Pythagorean theorem: $a^2 + b^2 = c^2$. Here, the sides are $AB = 8$ and $BC = 6$, and the hypotenuse is AC. So, $AC^2 = AB^2 + BC^2$. $AC^2 = 8^2 + 6^2$ $AC^2 = 64 + 36$ $AC^2 = 100$ $AC = \sqrt{100}$ $AC = 10$ units. Common mistakes include adding the sides (8 + 6 = 14) or multiplying them (8 * 6 = 48) instead of applying the Pythagorean theorem.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Formula and Secant Line Concept

State the formula for the average rate of change of $f(x)$ over $[a, b]$.

a. $\frac{f(b) - f(a)}{b - a}$ ✓

b. $\frac{f(a) + f(b)}{2}$

c. $\frac{f(b) - f(a)}{2}$

d. $\frac{b - a}{f(b) - f(a)}$

The average rate of change is the slope of the secant line passing through $(a, f(a))$ and $(b, f(b))$: $\frac{f(b) - f(a)}{b - a}$.

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22 [Foundational] • Target: Formula and Secant Line Concept

Find the average rate of change of $f(x) = x^2 - 3x$ from $x = 1$ to $x = 4$.

a. 2 ✓

b. 4

c. 3

d. 1

$f(1) = 1 - 3 = -2$. $f(4) = 16 - 12 = 4$. Rate $= \frac{4 - (-2)}{4 - 1} = \frac{6}{3} = 2$.

23 [Foundational] • Target: Formula and Secant Line Concept

Find the slope of the secant line connecting $(2, f(2))$ and $(5, f(5))$ for $f(x) = 2x^2 + 1$.

a. 14 ✓

b. 12

c. 16

d. 10

$f(2) = 9$, $f(5) = 51$. Slope $= \frac{51 - 9}{5 - 2} = \frac{42}{3} = 14$.

24 [Foundational] • Target: Tabular and Data-Based Rates of Change

A table shows $t = [0, 2, 5]$ and distance $s(t) = [0, 24, 75]$. What is the average velocity from $t = 2$ to $t = 5$?

a. 17 m/s ✓

b. 15 m/s

c. 25 m/s

d. 12 m/s

Average velocity $= \frac{s(5) - s(2)}{5 - 2} = \frac{75 - 24}{3} = \frac{51}{3} = 17$ m/s.

25 [Foundational] • Target: Tabular and Data-Based Rates of Change

Water depth in a reservoir drops from 80 ft at day 4 to 68 ft at day 10. Find the average rate of change in depth per day.

a. −2 ft/day ✓

b. 2 ft/day

c. −1.2 ft/day

d. −12 ft/day

$\frac{68 - 80}{10 - 4} = \frac{-12}{6} = -2$ ft/day.

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26 [Foundational] • Target: Tabular and Data-Based Rates of Change

If the average rate of change of $g(x)$ over $[1, 6]$ is 4 and $g(1) = 5$, find $g(6)$.

a. 25 ✓

b. 20

c. 29

d. 24

$\frac{g(6)-5}{6-1} = 4 \Rightarrow g(6) - 5 = 20 \Rightarrow g(6) = 25.$

27 [Foundational] • Target: Algebraic Expressions with Variables

Find the average rate of change of $f(x) = x^2$ over the interval $[2, 2 + h]$ in terms of h.

a. $4 + h$ ✓

b. 4

c. $2 + h$

d. $4 + 2h$

$\frac{f(2+h)-f(2)}{h} = \frac{(2+h)^2-4}{h} = \frac{4+4h+h^2-4}{h} = \frac{4h+h^2}{h} = 4 + h.$

28 [Foundational] • Target: Algebraic Expressions with Variables

Find the average rate of change of $f(x) = 3x^2 - 5$ over $[x, x + h]$.

a. $6x + 3h$ ✓

b. $6x$

c. $3x + 3h$

d. $6x - 5$

$\frac{3(x+h)^2-5-(3x^2-5)}{h} = \frac{6xh+3h^2}{h} = 6x + 3h.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Algebraic Expressions with Variables

For $f(x) = \frac{1}{x}$, find the average rate of change over $[a, b]$.

a. $-\frac{1}{ab}$ ✓

b. $\frac{1}{ab}$

c. $-\frac{1}{a+b}$

d. $\frac{a-b}{ab}$

$\frac{1/b-1/a}{b-a} = \frac{(a-b)/(ab)}{b-a} = -\frac{b-a}{ab(b-a)} = -\frac{1}{ab}.$

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30 [Intermediate] • Target: Contextual Applications

A projectile's height is given by $h(t) = -16t^2 + 80t$. Find its average velocity over the first 3 seconds $[0, 3]$.

a. 32 ft/s ✓

b. 48 ft/s

c. 16 ft/s

d. 80 ft/s

$h(0) = 0$. $h(3) = -16(9) + 80(3) = -144 + 240 = 96$ ft. Average velocity $= \frac{96 - 0}{3 - 0} = 32$ ft/s.

31 [Intermediate] • Target: Contextual Applications

A company's revenue in thousands of dollars is $R(x) = 50x - x^2$, where x is in hundreds of units. What is the average rate of change in revenue as production increases from $x = 10$ to $x = 20$?

a. 20 thousand dollars per unit ✓

b. 30

c. 15

d. 25

$R(10) = 500 - 100 = 400$. $R(20) = 1000 - 400 = 600$. Rate $= \frac{600 - 400}{20 - 10} = \frac{200}{10} = 20$.

32 [Intermediate] • Target: Contextual Applications

Why does a linear function $f(x) = mx + b$ have the same average rate of change over EVERY interval?

a. **Because the slope of the secant line between any two points is constantly m** ✓

b. Because $b = 0$

c. Because the function has no intercepts

d. Because $f(x)$ is symmetric

For any a and b, $\frac{(mb + b_0) - (ma + b_0)}{b - a} = \frac{m(b - a)}{b - a} = m$, which is constant.

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33 [Intermediate] • Target: Applications

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$ ✓

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1-x^2$

d. $\sqrt{x^2-1}$

Let $\theta = \sin^{-1}(x)$, so $\sin\theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where $\cos\theta \geq 0$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-x^2}$.

34 [Intermediate] • Target: Comparing rates and estimating from a table

For $0 < x < 1$, simplify $\sec\left(\tan^{-1}(x)\right)$ as an algebraic expression in x.

a. $1+x^2$

b. $\frac{1}{\sqrt{1+x^2}}$

c. $\sqrt{1-x^2}$

d. $\sqrt{1+x^2}$ ✓

Let $\theta = \tan^{-1}(x)$, so $\tan\theta = x$ with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, where $\sec\theta > 0$. Then $\sec\theta = \sqrt{1+\tan^2\theta} = \sqrt{1+x^2}$.

35 [Intermediate] • Target: Average rate of change and secant lines

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{5}\right)\right)$ as an algebraic expression in x.

a. $\sqrt{25-x^2}$

b. $\frac{\sqrt{25-x^2}}{5}$ ✓

c. $\frac{x}{\sqrt{25-x^2}}$

d. $\frac{\sqrt{25+x^2}}{5}$

Let $\theta = \sin^{-1}\left(\frac{x}{5}\right)$: opposite $= x$, hypotenuse $= 5$, adjacent $= \sqrt{25-x^2}$, and $\cos\theta \geq 0$. So $\cos\theta = \frac{\sqrt{25-x^2}}{5}$.

36 [Intermediate] • Target: Comparing average rate of change across different functions

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{2}\right)\right)$ as an algebraic expression in x.

a. $\sqrt{4-x^2}$

b. $\frac{\sqrt{4+x^2}}{2}$

c. $\frac{x}{\sqrt{4-x^2}}$

d. $\frac{\sqrt{4-x^2}}{2}$ ✓

Let $\theta = \sin^{-1}\left(\frac{x}{2}\right)$: opposite $= x$, hypotenuse $= 2$, adjacent $= \sqrt{4-x^2}$, and $\cos\theta \geq 0$. So $\cos\theta = \frac{\sqrt{4-x^2}}{2}$.

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37 [Intermediate] • Target: Average rate of change from a graph

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right)$ as an algebraic expression in x.

a. $\frac{\sqrt{9+x^2}}{3}$

b. $\sqrt{9-x^2}$

c. $\frac{\sqrt{9-x^2}}{3}$ ✓

d. $\frac{x}{\sqrt{9-x^2}}$

Let $\theta = \sin^{-1}\left(\frac{x}{3}\right)$: opposite $= x$, hypotenuse $= 3$, adjacent $= \sqrt{9-x^2}$, and $\cos\theta \geq 0$. So $\cos\theta = \frac{\sqrt{9-x^2}}{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 7.02 Logarithms and Their Properties

For $0 < x < 1$, simplify $\sin\left(\tan^{-1}\left(\frac{x}{4}\right)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{x^2+16}}$ ✓

b. $\frac{x}{4}$

c. $\frac{x}{\sqrt{16-x^2}}$

d. $\frac{4}{\sqrt{x^2+16}}$

Let $\theta = \tan^{-1}\left(\frac{x}{4}\right)$: opposite $= x$, adjacent $= 4$, hypotenuse $= \sqrt{x^2+16}$. So $\sin\theta = \frac{x}{\sqrt{x^2+16}}$.

39 [Challenging] • Target: Formula and Secant Line Concept • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

For $0 < x < 1$, simplify $\sin\left(\tan^{-1}\left(\frac{x}{3}\right)\right)$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{9-x^2}}$

b. $\frac{x}{3}$

c. $\frac{x}{\sqrt{x^2+9}}$ ✓

d. $\frac{3}{\sqrt{x^2+9}}$

Let $\theta = \tan^{-1}\left(\frac{x}{3}\right)$: opposite $= x$, adjacent $= 3$, hypotenuse $= \sqrt{x^2+9}$. So $\sin\theta = \frac{x}{\sqrt{x^2+9}}$.

40 [Challenging] • Target: Tabular and Data-Based Rates of Change • Cross-Lesson: 7.04 Applications: Growth and Decay

For $0 < x < 1$, simplify $\sin\left(\cos^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1+x^2}$

b. $-\sqrt{1-x^2}$

c. $1-x$

d. $\sqrt{1-x^2}$ ✓

Let $\theta = \cos^{-1}(x)$, so $\cos\theta = x$ and $\theta \in [0, \pi]$, where $\sin\theta \geq 0$. Then $\sin\theta = \sqrt{1-x^2}$.

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41 [Challenging] • Target: Algebraic Expressions with Variables • Cross-Lesson: 8.01 Rational Functions and Asymptotes

For $0 < x < 1$, simplify $\tan(\sin^{-1}(x))$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{1+x^2}}$

b. $\frac{x}{1-x^2}$

c. $\frac{x}{\sqrt{1-x^2}}$ ✓

d. $\frac{\sqrt{1-x^2}}{x}$

Let $\theta = \sin^{-1}(x)$: opposite $= x$, hypotenuse $= 1$, adjacent $= \sqrt{1-x^2}$. So $\tan\theta = \frac{x}{\sqrt{1-x^2}}$.

42 [Challenging] • Target: Contextual Applications • Cross-Lesson: 8.02 Graphing Rational Functions

For $0 < x < 1$, simplify $\tan(\cos^{-1}(x))$ as an algebraic expression in x.

a. $\frac{\sqrt{1-x^2}}{x}$ ✓

b. $\frac{1-x^2}{x}$

c. $\frac{x}{\sqrt{1-x^2}}$

d. $\frac{\sqrt{1+x^2}}{x}$

Let $\theta = \cos^{-1}(x)$: adjacent $= x$, hypotenuse $= 1$, opposite $= \sqrt{1-x^2}$. So $\tan\theta = \frac{\sqrt{1-x^2}}{x}$.

43 [Challenging] • Target: Computing average rate of change • Cross-Lesson: 8.03 Introduction to Limits

For $0 < x < 1$, simplify $\sin(\tan^{-1}(x))$ as an algebraic expression in x.

a. $\frac{x}{\sqrt{1+x^2}}$ ✓

b. $\frac{x}{1+x^2}$

c. $\frac{1}{\sqrt{1+x^2}}$

d. $\frac{x}{\sqrt{1-x^2}}$

Let $\theta = \tan^{-1}(x)$: opposite $= x$, adjacent $= 1$, hypotenuse $= \sqrt{1+x^2}$. So $\sin\theta = \frac{x}{\sqrt{1+x^2}}$.

44 [Challenging] • Target: Applications • Cross-Lesson: 9.01 Arithmetic Sequences and Series

For $0 < x < 1$, simplify $\cos(\tan^{-1}(x))$ as an algebraic expression in x.

a. $\frac{1}{\sqrt{1+x^2}}$ ✓

b. $\sqrt{1+x^2}$

c. $\frac{1}{\sqrt{1-x^2}}$

d. $\frac{x}{\sqrt{1+x^2}}$

Let $\theta = \tan^{-1}(x)$: opposite $= x$, adjacent $= 1$, hypotenuse $= \sqrt{1+x^2}$. So $\cos\theta = \frac{1}{\sqrt{1+x^2}}$.

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45 [Challenging] • Target: Comparing rates and estimating from a table • Cross-Lesson: 9.02 Geometric Sequences and Series

For $0 < x < 1$, simplify $\sin\left(2\cos^{-1}(x)\right)$ as an algebraic expression in x.

a. $2x\sqrt{1-x^2}$ ✓

b. $2x^2 - 1$

c. $2\sqrt{1-x^2}$

d. $x\sqrt{1-x^2}$

Let $\theta = \cos^{-1}(x)$, so $\sin\theta = \sqrt{1-x^2}$. Then $\sin 2\theta = 2\sin\theta\cos\theta = 2x\sqrt{1-x^2}$.

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